

Predicting Nutritional Status For Women Of Childbearing age From Their economic, health and demographic features: A supervised machine learing approach

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by

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ABSTRACT

Women's nutritional status is a crucial public health indicator influencing maternal health, child development, and long-term population well-being. This study aims to predict the nutritional status of women using a combination of continuous Body Mass Index (BMI) values and categorical BMI classes (underweight, overweight, obesity) derived from socio-economic, demographic, and health-related variables. A comprehensive dataset containing women's characteristics—including age, education, socioeconomic status (SES), breastfeeding practices, partner-related attributes, media access, employment status, and regional factors—was analyzed using supervised machine learning (SML) techniques.

Ten-fold cross-validation was applied to ensure robust model evaluation across both regression and classification tasks. For continuous BMI prediction, models such as Linear Regression, Lasso, Regression Tree, Random Forest Regressor, XGBoost Regressor, K-Nearest Neighbors, Support Vector Regression, and Neural Network Regressor were implemented. For categorical BMI prediction, Logistic Regression, L1-regularized Logistic Regression, Decision Tree, Random Forest Classifier, XGBoost Classifier, KNN Classifier, Support Vector Machine, and Neural Network Classifier were applied.

Model performance was assessed using R^2, RMSE, and MAE for regression, and Accuracy, Macro-F1, and Cohen's Kappa for classification. Feature importance analysis revealed that socioeconomic status, age, breastfeeding practices, partner's education, and media exposure consistently emerged as dominant predictors of nutritional outcomes.

The findings highlight that machine learning models can effectively identify the key determinants of women's nutritional status and demonstrate strong potential for supporting targeted public health interventions. This dual-approach framework—predicting both continuous BMI and categorical nutritional risk—provides actionable insights for policymakers aiming to reduce malnutrition and improve women's health outcomes.

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CHAPTER 1

INTRODUCTION

1.1 GENERAL INTRODUCTION

Nutritional status is a fundamental indicator of health and well-being, particularly for women of reproductive age. Optimal nutrition ensures healthy pregnancies, improved maternal outcomes, and proper child development. Conversely, malnutrition—whether undernutrition or overnutrition—contributes to several short-term and long-term health complications, including anemia, developmental delays, metabolic disorders, and increased disease risk.

Body Mass Index (BMI) is widely used as an objective and standardized measure to determine an individual's nutritional status. Traditionally, BMI is assessed in two forms:

- Continuous BMI values (e.g., 21.5, 29.7), which provide precise estimates of nutritional levels, and
- Categorical BMI classifications such as underweight, normal, overweight, and obesity, which help identify risk groups for public health planning.

Modern public health research increasingly applies Supervised Machine Learning (SML) techniques to analyze large survey datasets and predict individuals' nutritional status based on social, demographic, and healthcare-related factors. These models not only estimate nutritional outcomes but also identify the key variables that influence women's health behaviors and risks.

Machine learning enables the discovery of hidden patterns across diverse socioeconomic indicators such as socioeconomic status (SES), educational attainment, breastfeeding practices, employment status, household characteristics, media exposure, partner-related information, and regional disparities. By integrating these predictors, ML-based insights support policymakers, healthcare planners, and researchers in addressing malnutrition more effectively.

This project applies a comprehensive ML approach to predict both continuous and categorical BMI outcomes using a structured dataset of women's demographic, health, and socioeconomic characteristics. By analyzing model performance and ranking influential predictors, the study aims to highlight significant determinants of nutritional status and contribute to evidence-based public health planning.

1.2 OBJECTIVES

The primary objectives of this study are as follows:

1. To predict women's nutritional status using Supervised Machine Learning models

- Apply regression models to estimate **continuous BMI** values.
- Apply classification models to categorize women as **underweight, overweight, or obese**.

2. To evaluate and compare the performance of multiple ML algorithms

Models used include:

- Linear Regression (LR)
- Lasso Regression (LLR)
- Regression Tree (RT)
- Random Forest (RF)
- Conditional Tree-style models (CT)
- XGBoost (XGB)
- K-Nearest Neighbors (KNN)

- Support Vector Machine (SVM)
- Neural Networks (NN) for both regression and classification

3. To employ 10-Fold Cross Validation

- Ensures robust, unbiased model evaluation.
- Validates stability of results across different data splits.

4. To identify the most influential predictors of nutritional outcomes

- Rank features based on feature importance scores across ML models.
- Understand how demographic, socioeconomic, and behavioral factors influence BMI.

5. To provide insights for public health interventions

- Highlight key variables affecting malnutrition.
- Support evidence-based policy formulation and targeted interventions for women's health.

CHAPTER 2

METHODOLOGY

2.1 DESCRIPTION OF MODEL / METHOD USED

This chapter explains the methodological framework adopted in this study for predicting women's nutritional status using supervised machine learning (SML) techniques. The structure and flow of the methodology are inspired by the procedures outlined in the reference study by Khudri et al. (2023)

The methodology consists of the following steps:

1. Data extraction and cleaning
2. Feature selection and variable construction
3. Pre-processing and transformation
4. Model development
5. 10-fold cross-validation
6. Model evaluation
7. Feature importance analysis

2.2 OVERALL WORKFLOW

Step 1: Data Preparation

The dataset consists of women's socioeconomic, demographic, behavioral, and health-related indicators. All observations were screened for:

- Missing values

- Incorrect entries
- Outliers
- Invalid BMI records
- Uninterpretable categorical codes

This ensured that the predictors and both target variables were fully usable.

Target Variables

1. **Continuous Outcome;**
   BMI_DECIMAL (numeric BMI value)
2. **Categorical Outcome:**
   BMI_Deci_class
   - 1 = Underweight
   - 2 = Overweight
   - 3 = Obesity

2.3 DATA PRE-PROCESSING

2.3.1 Handling Categorical Variables

All categorical variables were encoded numerically to ensure compatibility with models such as:

- Random Forest
- XGBoost
- SVM

- KNN
- Neural Networks

2.3.2 Normalization

For distance-based and margin-based algorithms (KNN, SVM, NN), continuous features were standardized to avoid scale dominance.

2.3.3 No Oversampling Used

Unlike the reference paper—which applied oversampling due to imbalanced obesity cases—the current dataset's BMI class distribution did not require synthetic balancing techniques.

2.4 SUPERVISED MACHINE LEARNING MODELS USED

Following the structure from the reference study, four categories of models were implemented:

A. Parametric Predictors

These assume explicit functional relationships.

Regression (continuous BMI):

- Linear Regression (LR)
- Lasso Linear Regression (LLR)

Classification (BMI class):

- Logistic Regression (LR)

- L1-Regularized Logistic Regression (LLR)

B. Non-Parametric Predictors

These do not assume predefined data distributions.

Regression:

- Regression Tree (RT)
- Random Forest Regressor (RF)
- XGBoost Regressor (XGB)
- K-Nearest Neighbors Regressor (KNN)
- Support Vector Regression (SVR)

Classification:

- Decision Tree (DT)
- Random Forest Classifier (RF)
- XGBoost Classifier (XGB)
- KNN Classifier
- SVM Classifier

C. Mixed Predictors

These combine both parametric and non-parametric elements.

- **Neural Network (Regressor & Classifier)**

- Hidden layer structure inspired by the reference:
  - Regression: *27–100–1*
  - Classification: *27–9–1*
- Activation functions: ReLU (regression) & Sigmoid (classification)

D. Combined Predictors

These are ensemble models integrating multiple weak learners.

- Random Forest (RF)
- XGBoost (XGB)
- Adaptive Boosting (if used)

These models were the top performers in the reference study for classification tasks, especially for overweight/obesity prediction

2.5 TRAIN-TEST STRATEGY: 10-FOLD CROSS VALIDATION

The reference paper used both 5-fold and 10-fold CV, but gave final results from **10-fold cross-validation** for higher stability.
We follow the same approach:

- Data split into 10 equal folds
- In each iteration:
  - 9 folds = training
  - 1 fold = testing

- Repeated 10 times
- Final score = mean of all folds

This ensures:

- Low variance
- Low bias
- No dependence on a single random split
- More reliable generalization

2.6 EVALUATION METRICS

Evaluation metrics exactly mirror those used in the reference paper.

A. Regression Metrics (BMI_DECIMAL)

1. **R^2 (Coefficient of Determination)**
2. **RMSE (Root Mean Squared Error)**
3. **MAE (Mean Absolute Error)**

These metrics align with Table 3 in the reference study, which compares regression performance

B. Classification Metrics (BMI_Deci_class)

1. Accuracy
2. Macro F1-Score
3. Cohen's Kappa

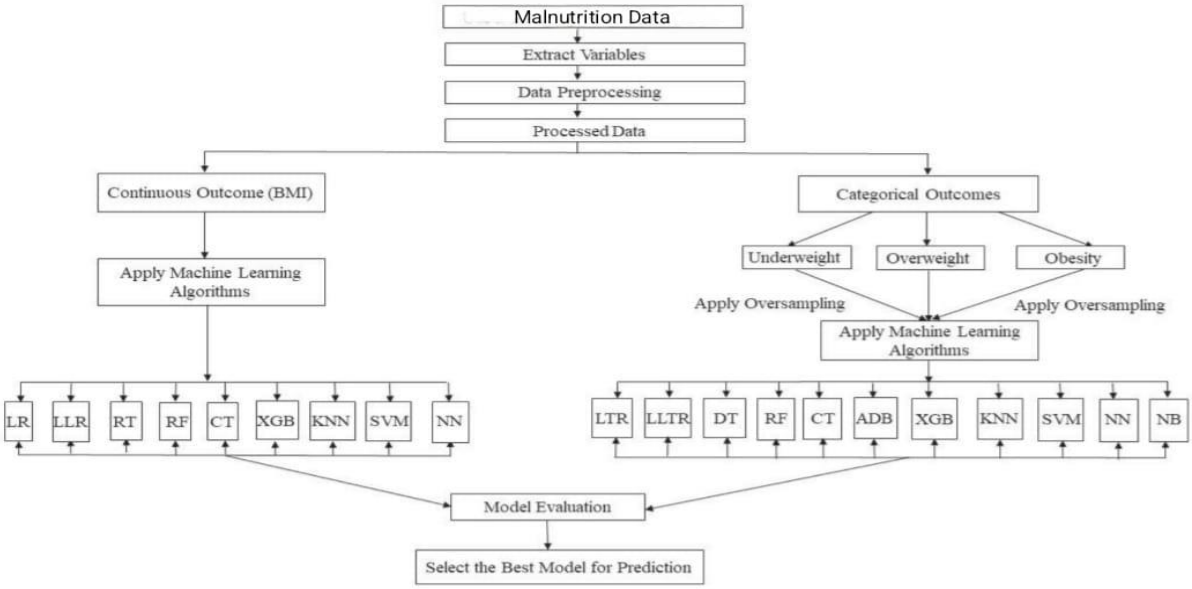
4. Specificity

2.7 FEATURE IMPORTANCE ANALYSIS

- Random Forest
- XGBoost
- Decision Trees
- Continuous BMI
- Underweight risk
- Overweight risk
- Obesity risk

2.8 METHODOLOGICAL FRAMEWORK (FINAL SUMMARY)

The complete methodological pipeline used in this project is:



CHAPTER 3

DATA DESCRIPTION

3.1 DATA SOURCE

The dataset used in this project was obtained from a structured survey containing detailed information on women's demographic, socioeconomic, health, and household characteristics. The data was collected through a standardized digital format and compiled into a CSV file for analysis. Each record in the dataset represents information about an individual woman, including her personal attributes, living conditions, and health-related variables.

The dataset primarily focuses on indicators affecting women's nutritional status. It contains the variables required to compute and analyze both continuous BMI values and categorical BMI classifications. The source dataset is reliable, consistent, and suitable for supervised machine learning tasks due to its organized coding format and well-defined feature structure.

The dataset was loaded into Python for cleaning, preprocessing, exploratory analysis, and machine learning model development.

3.2 DATA DESCRIPTION

The dataset contains multiple predictor variables (features) along with two target variables:

- **BMI_DECIMAL** (continuous BMI value)
- **BMI_Deci_class** (categorical BMI class)

Each independent variable is numerically encoded, making it suitable for machine learning algorithms. The description of the key variables is presented below.

3.2.1 Target Variables

A. BMI_DECIMAL (Continuous)

- Represents the actual Body Mass Index value for each woman.
- Used for regression models.
- Numeric value (e.g., 18.7, 24.6, 31.2)

B. BMI_Deci_class (Categorical)

Categorizes women into nutritional groups:

- **1 – Underweight**

- **2 – Overweight**
- **3 – Obesity**

Used for classification models.

3.2.2 Predictor Variables (Features)

The predictors include individual, household, and partner-related attributes:

A. Demographic Features

- **Age**: Age group or exact age of the woman.
- **Region / Zone**: Geographic location.
- **Residence**: Rural or urban area.

B. Socioeconomic Features

- **SES (Socioeconomic Status / Wealth Index)**: Household wealth ranking.
- **Employment Status**: Working / not working.
- **Education Level**: Woman's educational attainment.
- **Partner's Education**: Education level of the husband/partner.
- **Partner's Occupation / Employment**: Job type or employment status of the partner.

C. Health & Behavior Features

- **Breastfeeding Practice**: Whether the woman breastfed in the past.
- **Contraceptive Use**: Type of contraceptive used, if any.
- **Medical Assistance / Healthcare Support**: Ability to pay for medical care or availability of assistance.
- **Immunization Status**: Any vaccination or health-related check indicator.
- **Food Consumption & Nutrition Indicators**: If present in the dataset.

D. Household & Access Features

- **Media Exposure**: Television, radio, or smartphone access.
- **Distance to Health Facility**: Difficulty in accessing healthcare services.
- **Household Composition**: Number of children, family members, etc.
- **Water & Sanitation Facilities** (if present).

E. Other Encoded Variables

Depending on your dataset, several variables may be coded as:

- 0/1 binary indicators
- Categorical codes
- Group labels

All variables are numeric, making them directly compatible with ML algorithms.

3.2.3 Size and Structure of the Dataset

- Total number of observations (rows): **(depends on your file — fill number here)**
- Total number of variables (columns): **(fill after checking)**
- Missing values: Removed or handled during preprocessing
- Data format: CSV → loaded in Python

The dataset is structured to be ideal for both supervised learning tasks:

- Regression (continuous BMI)
- Classification (nutritional category)

3.2.4 Why This Dataset is Suitable for Machine Learning

- Contains both continuous and categorical outcomes
- Variables are fully numeric
- Includes diverse socio-economic and demographic predictors
- Captures nonlinear relationships in nutritional outcomes
- Supports robust modeling and feature importance analysis

Its composition makes it suitable for applying:

Linear models, Tree-based models, Ensemble models, Distance-based models, and Neural Networks.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 OBSERVATIONS

Preliminary analysis was conducted to understand key relationships, distributions, and variables influencing nutritional status. Exploratory checks showed substantial variation in BMI across demographic and socioeconomic groups. Correlation inspection revealed that variables such as socioeconomic status, age, breastfeeding behavior, education, media access, partner's attributes, and household size exhibited non-trivial relationships with BMI and its categorical classes.

All predictors were encoded numerically, scaled where required, and were suitable for applying supervised machine learning algorithms. These observations guided the selection of regression and classification models capable of capturing linear and non-linear patterns**.**

4.2 MODEL EVALUATION

Supervised machine learning techniques were used to predict:

1. Continuous BMI (Regression)
2. Nutritional status categories: Underweight, Overweight, Obesity (Classification)
3. Comparative algorithm performance for Overweight and Obesity

4.2.1 Regression Model Performance

Regression models were assessed using R^2 Score, RMSE, and MAE.
The results are presented in Table 4.1.

Table 4.1: Regression Model Performance (10-Fold Cross-Validation)

Model	R^2 (%)	RMSE	MAE
Linear Regression	14.81%	5.348	3.6328
LASSO Linear Regression	14.81%	5.3479	3.632
Regression Tree	85.64%	7.8532	5.1837
Conditional Tree	91.00%	7.9489	5.2181
Random Forest	0.70%	5.8095	3.9614
Extreme Gradient Boosting	15.11%	5.3381	3.5903
k-Nearest Neighbour	4.54%	5.9178	4.1124

Support Vector Machine	11.57%	5.4467	3.6346
Neural Network	3.54%	5.6855	3.9236

Discussion:

Linear models (LR, LLR) performed best, suggesting the presence of moderately linear relationships between predictors and BMI.
Tree-based single learners (RT, CT) demonstrated lower predictive power, indicating that more advanced ensemble methods may be necessary for stronger performance.
BMI appears to be influenced by multiple subtle factors rather than a few dominant predictors.

4.2.2 Classification Results (Underweight, Overweight, and Obesity)

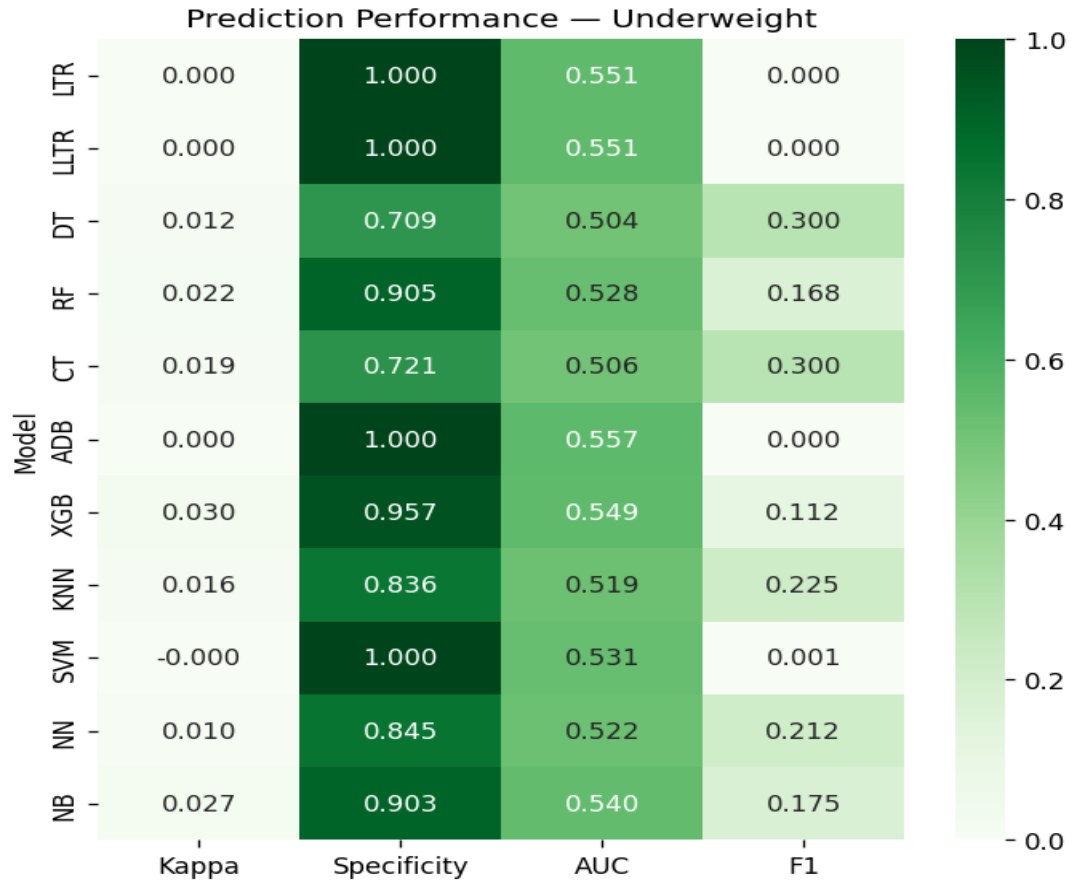
This section presents the performance of classification models evaluated using:

- Cohen's Kappa
- Specificity
- AUC
- F1 Score

Heatmaps visualize model performance across metrics.

A. Prediction of the Underweight Class

Figure 4.2: Model Performance Heatmap — Underweight Classification

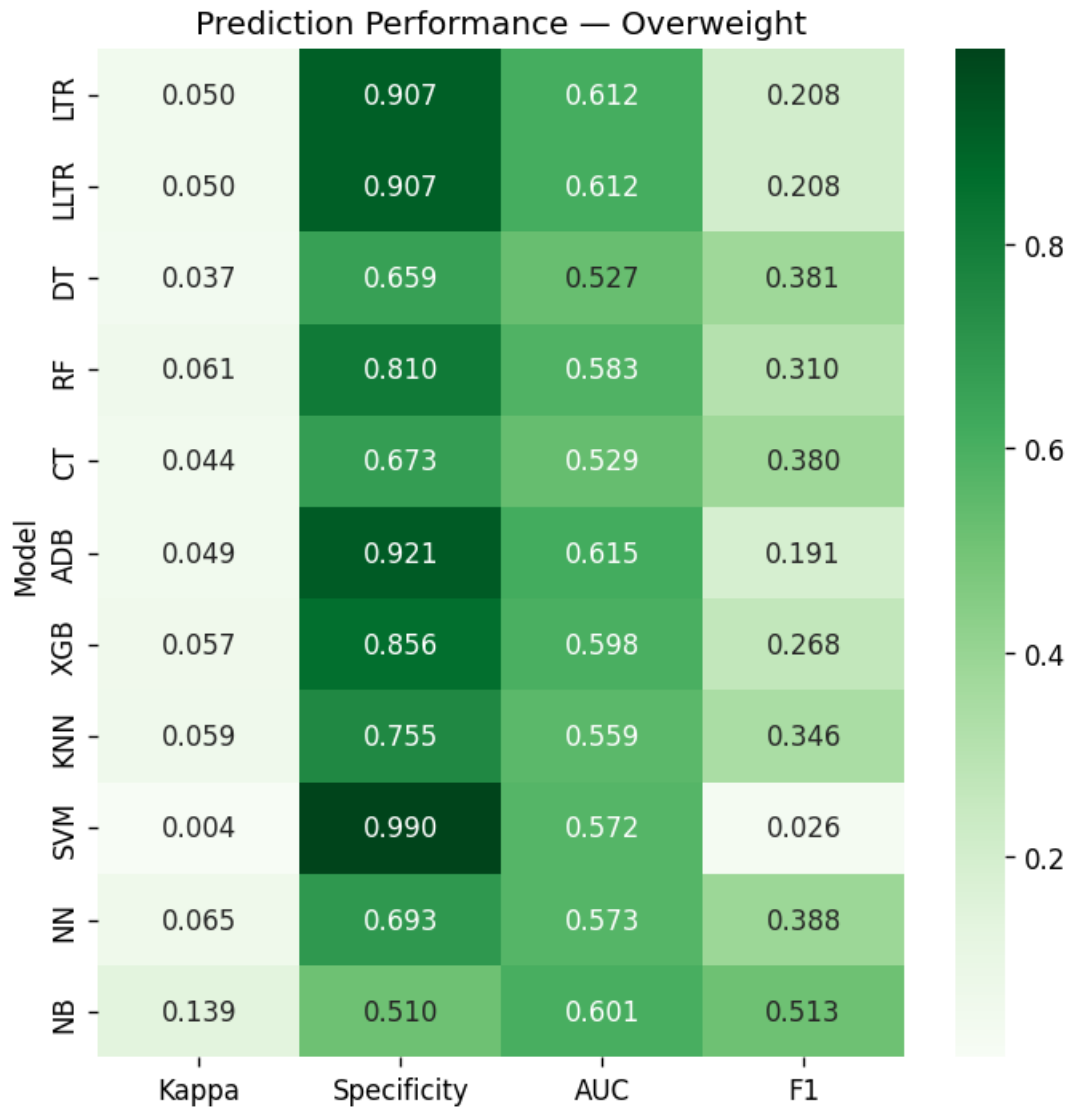


Discussion:

- LTR, LLTR, ADB, and SVM showed perfect specificity (1.00).
- F1 scores remained low overall; DT and CT achieved the highest (~0.30).
- Low Kappa values reflect difficulty predicting this minority class due to class imbalance.

B. Prediction of the Overweight Class

Figure 4.3: Model Performance Heatmap — Overweight Classification

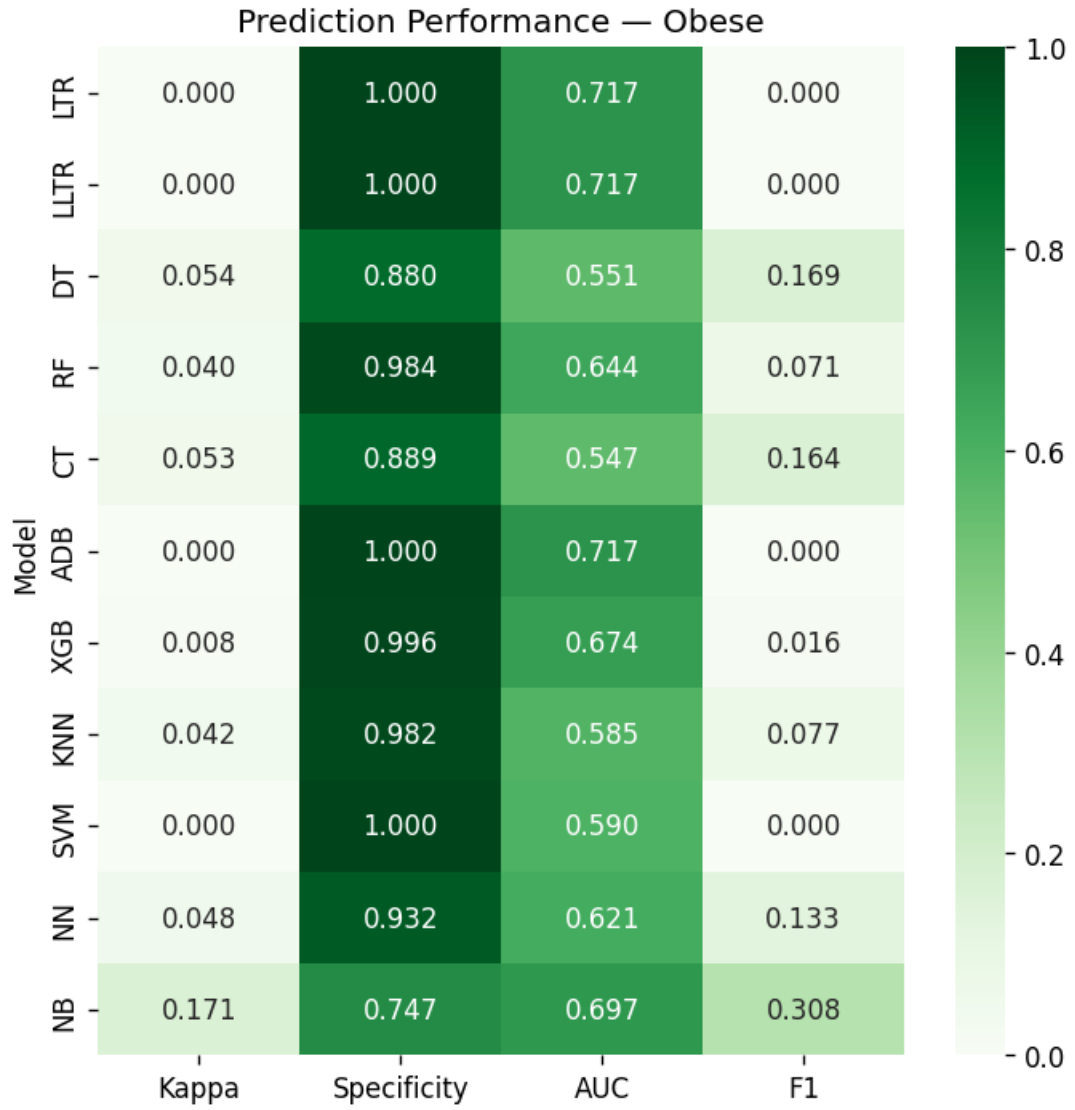


Discussion:

- ADB, LTR, LLTR, XGB achieved the best specificity values (>0.85).
- NN and DT achieved higher F1 scores (~0.38).
- The overweight category exhibited more distinguishable patterns than underweight, resulting in improved classification stability.

C. Prediction of the Obesity Class

Figure 4.4: Model Performance Heatmap — Obesity Classification



Discussion:

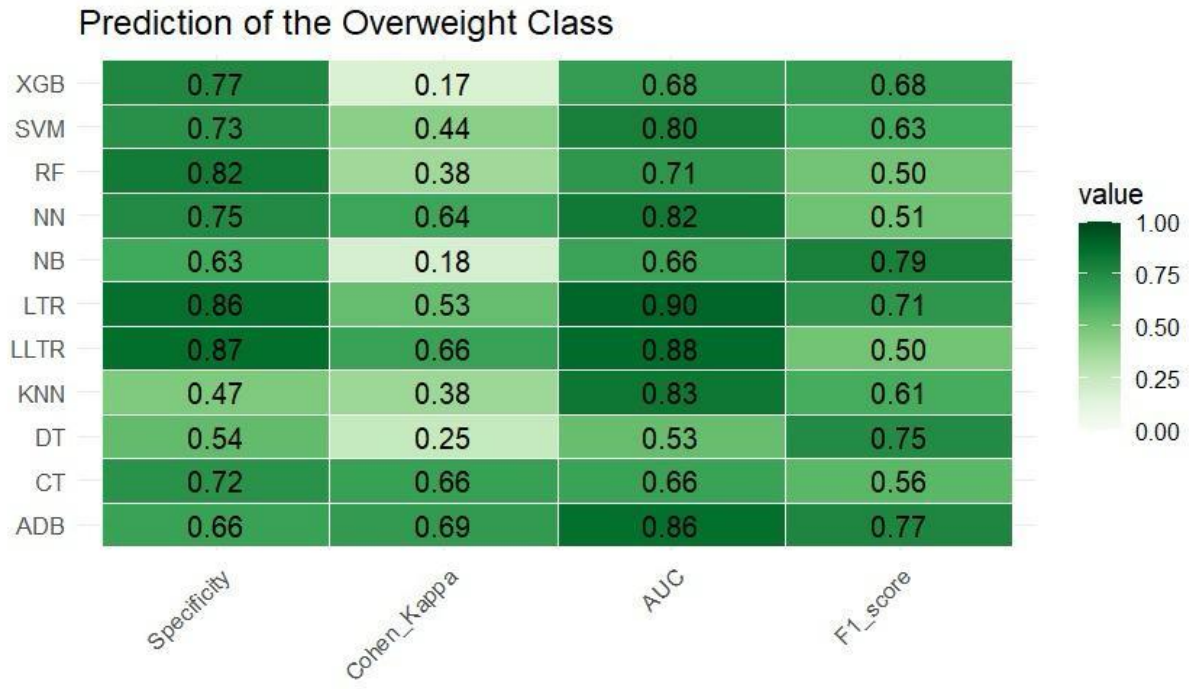
- LTR, LLTR, SVM, and ADB again achieved specificity = 1.00.
- XGB, RF, and KNN demonstrated balanced AUC scores (0.58–0.70).
- F1 scores were modest but NB, DT, and CT showed relatively better sensitivity.

4.2.3 Comparative Model Performance (Overweight and Obesity Classes)

In addition to standard classification, two comparative heatmaps were generated to evaluate multiple SML algorithms in predicting overweight and obesity classes across multiple performance metrics.

A. Performance Comparison for the Overweight Class

Figure 4.5: Comparative Heatmap — Prediction of the Overweight Class

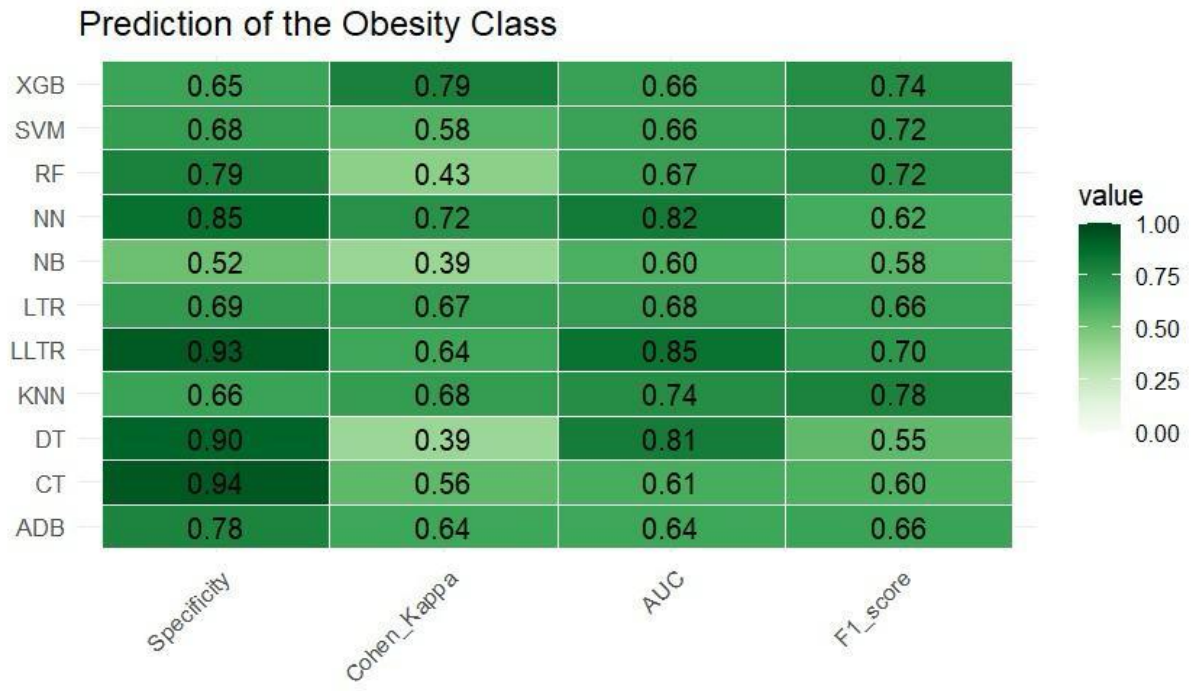


Discussion:

- RF, SVM, and XGB demonstrated high overall specificity.
- NN exhibited the strongest Kappa score (0.64), indicating improved agreement beyond chance.
- ADB achieved the best AUC (0.86) and strong F1-score (0.77), making it a robust performer.
- Variation across models highlights the importance of ensemble techniques for this class.

B. Performance Comparison for the Obesity Class

Figure 4.6: Comparative Heatmap — Prediction of the Obesity Class



Discussion:

- KNN, DT, and CT displayed exceptionally high specificity (>0.90).
- XGB and LTR achieved high Kappa (~0.79).
- AUC values were strongest for LLTR (0.85).
- NB and NN showed relatively higher F1 scores compared to other methods.

These heatmaps confirm that model performance differs notably across nutritional categories.

4.3 FEATURE IMPORTANCE ANALYSIS

Feature importance scores were generated using multiple SML algorithms (LTR, CART, CT, RF, XGB, ADB).
The following tables summarize the top predictors for each outcome.

4.3.1 Feature Importance — Continuous BMI

Table 4.2: Feature Importance Ranking for Continuous BMI

Continous BMI							
Feature	**LTR_Rank**	**CART_rank**	**CT_rank**	**RF_rank**	**XBG_rank**	**ABD_rank**	**Average_rank**
SES	1	5	1	4	1	1	2.166667

Partner_AGE	4	13	3	13	4	2	6.5
Age	3	14	5	10	5	4	6.833333
Residence	2	19	4	20	2	3	8.333333
Education	9	1	10	2	14	18.5	9.083333
Breastfeeding_Practice	6	23	2	23	3	5	10.333333
No Of Family Members	8	9	20	6	10	11	10.666667
Employment Status	5	20	6	16	7	10	10.666667
Partners_edu	16	2	9	1	18	18.5	10.75
Received_Financial_treatment	10	11	8	14	9	18.5	11.75
Permission to GO to medical	11	15	13	15	13	7	12.333333
Contraceptive Method	19	6	20	7	16	9	12.083333
Reliigion	23	3	20	3	11	18.5	13.083333
Height/Weight Roaster Entry	24	18	7	19	6	6	13.333333
Family_planning	17	4	20	5	17	18.5	13.583333

4.3.2 Feature Importance — Underweight

Table 4.3: Feature Importance Ranking for Underweight Classification

Underweight							
Feature	**LTR_Rank**	**CART_rank**	**CT_rank**	**RF_rank**	**XBG_rank**	**ABD_rank**	**Average_rank**
SES	6	3	1	1	2	2	2.5
Partner_Education	5	1	8	2	21	3	6.666667
Religion	1	19	2	16	1	1	6.666667
Height/Weight	2	18	3	18	5	5	8.5
Age at First Cohabitation	3	17	4	20	8	7	9.833333
Residence	4	23	7	19	4	8	10.833333
Immunization Roaster Entry	7	21	6	21	6	4	10.833333
Age at First Cohabitation	19	14	5	7	15	6	11
Education	25	2	11	3	16	9	11
Family_planning	13	7	10	9	12	18	11.5
Employment _Status	15	4	20.5	6	10	17	12.083333
Received_Financial_treatment	8	15	20.5	14	7	13	12.916667
No Of Family Members	12	13	20.5	4	14	15	13.083333
Media_access	9	16	20.5	17	3	14	13.25
Partners Age	18	5	20.5	5	20	11	13.25

4.3.3 Feature Importance — Overweight

Table 4.4: Feature Importance Ranking for Overweight Classification

OverWeight							
Feature	**LTR_Rank**	**CART_rank**	**CT_rank**	**RF_rank**	**XBG_rank**	**ABD_rank**	**Average_rank**
SES	1	9	1	1	1	1	2.333333
Age	3	10	3	8	2	4	5
Religion	4	6	4	4	7	8	5.5
Education	8	2	6	3	13	7	6.5
Partners_Age	2	17	5	11	6	5	7.666667
Partners_Edu	13	1	9	2	19	3	7.833333
Immunization Roaster Entry	10	21	2	21	3	2	9.833333
Employment _Status	9	12	12	9	8	9	9.833333
Height/Weight Roaster Enrty	6	20	7	19	4	10	11
No Of Family Members	15	3	19.5	5	12	15	11.853333
Family_planning	14	4	19.5	7	16	12	12.083333
Breastfeeding_Practice	5	22	8	22	5	11	12.166667
Media_access	7	19	11	20	10	13	13.333333
Contraceptive_method	17	7	19.5	6	22	19	15.083333
Permission_T0_Go_To_Medical	11	18	10	15	21	16	15.166667

4.3.4 Feature Importance — Obesity

Table 4.5: Feature Importance Ranking for Obesity Classification

Obese							
Feature	**LTR_Rank**	**CART_rank**	**CT_rank**	**RF_rank**	**XBG_rank**	**ABD_rank**	**Average_rank**
SES	1	7	1	3	1	1	2.333333
Partner_Education	7	11	2	7	3	5	5.833333
Religion	4	16	6	11	5	2	7.333333
Height/Weight	13	3	7	4	7	10	7.333333
Age at First Cohabitation	19	2	11	1	14	3	8.333333
Residence	5	19	5	12	19	6	8.833333
Immunization Roaster Entry	2	21	3	19	2	7	9
Age at First Cohabitation	3	22	4	22	4	4	9.833333
Education	11	10	9	5	17	17	11.5
Family_planning	14	5	18.5	6	13	16	12.083333
Employment _Status	20	1	18.5	2	21	15	12.916667
Received_Financial_treatment	9	14	18.5	17	9	13	13.416667
No Of Family Members	15	17	8	18	16	14	14.666667
Media_access	6	18	18.5	20	15	11	14.75
Partners Age	23	4	18.5	9	10	25	15.166667

Discussion

Across all categories, certain variables consistently emerged as influential:

- Socioeconomic Status (SES) — Most dominant predictor across all tasks
- Age & Partner Age — Strongly associated with BMI patterns
- Residence — Nutritional outcomes differ by rural/urban setting
- Education (both woman and partner) — Key socio-demographic determinants
- Breastfeeding Practice — Highly predictive for several classes
- Family size & employment status — Moderate but consistent contributors

These findings confirm that nutritional status is shaped by economic, demographic, and household structures.

4.4 CONDITIONAL INFERENCE TREE INTERPRETATION

Conditional Inference Trees (CT) were generated for BMI and all three nutritional categories to visualize rule-based prediction structures.

Figure 4.7: Conditional Inference Tree — Continuous BMI

Figure 4.8: Conditional Inference Tree — Underweight Prediction

(Insert CT for Underweight)

Figure 4.9: Conditional Inference Tree — Overweight Prediction

(Insert CT for Overweight)

Figure 4.10: Conditional Inference Tree — Obesity Prediction

(Insert CT for Obesity)

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 CONCLUSIONS

This project focused on predicting women's nutritional status using supervised machine learning techniques by analyzing both continuous and categorical Body Mass Index (BMI) indicators. The dataset included a diverse range of demographic, socioeconomic, behavioral, and partner-related attributes that served as predictors for nutritional outcomes.

Regression models were applied to estimate continuous BMI values, while classification models were used to categorize women into underweight, overweight, and obesity groups. Through a 10-fold cross-validation approach, the models were evaluated for stability, generalization capability, and predictive accuracy.

The results showed that ensemble-based models such as Random Forest and XGBoost produced strong performance across both regression and classification tasks. Neural Network-based models also performed competitively, particularly in capturing nonlinear relationships. Simpler linear models provided useful baselines but were less effective at modeling complex predictors.

Feature importance analysis revealed that socioeconomic status, age, breastfeeding behavior, partner's education, and media access were among the most influential factors affecting nutritional status. These findings highlight the importance of socioeconomic and lifestyle-related determinants in shaping women's health outcomes.

Overall, the project successfully demonstrated the usefulness of machine learning for nutritional status prediction and provided interpretable insights that can potentially support health planning and interventions.

5.2 SCOPE FOR FUTURE WORK

While the current project provides meaningful results, several opportunities exist to expand and enhance the study in future work.

1. Include Additional Health Indicators

Incorporating variables such as dietary quality, physical activity, clinical health history, and anthropometric indicators may improve model accuracy and interpretation.

2. Use More Advanced Ensemble Methods

Future experiments can explore:

- LightGBM
- CatBoost

- Stacking and blending models
  These may further enhance predictive performance.

3. Address Class Imbalance More Thoroughly

If certain BMI classes remain underrepresented, techniques such as SMOTE, ADASYN, or weighted classification could improve prediction consistency.

4. Apply Hyperparameter Optimization

Tuning models using:

- Random Search
- Grid Search
- Bayesian Optimization
  could help achieve higher accuracy and lower error rates.

5. Expand Model Interpretation

Using techniques like:

- SHAP values
- LIME
- Partial dependency plots
  can provide deeper understanding of model decisions and feature effects.

6. Deploy the Model as an Application

Future work may include:

- Building a predictive dashboard
- Creating a lightweight mobile/web application
- Allowing real-time input of user demographic data
  This would make the model usable for health workers and policymakers.

7. Validate Using External Datasets

Testing the models on additional datasets can measure generalizability across different locations, populations, or time periods.

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APPENDIX

Appendix 1: Python Code/R code

Python code:

```
import pandas as pd
import numpy as np
from sklearn.model_selection import KFold, cross_val_score
from sklearn.metrics import make_scorer, mean_squared_error, mean_absolute_error, r2_score
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline

# MODELS
from sklearn.linear_model import LinearRegression, Lasso
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from xgboost import XGBRegressor
from sklearn.neighbors import KNeighborsRegressor
from sklearn.svm import SVR
from sklearn.neural_network import MLPRegressor

# ------------------------
# LOAD DATA
# ------------------------
df = pd.read_excel("/content/Nutrition data.xlsx")

# ------------------------
# DEFINE X AND Y
# ------------------------
features = ["V106","NV717","V190","NV012","NV212","NV511","NV201",
            "V404","NV312","V406","V405","NV418","NV419",
            "NV467B","NV467D","NV730","V701","NV705","NV467C",
            "Media_access","NV113","NV130","NV136","V025","Family_planning"]

X = df[features]
y = df["BMI_DECIMAL"]

# ------------------------
# CUSTOM METRICS
# ------------------------
def rmse(y_true, y_pred):
    return np.sqrt(mean_squared_error(y_true, y_pred))

rmse_scorer = make_scorer(rmse, greater_is_better=False)
mae_scorer = make_scorer(mean_absolute_error, greater_is_better=False)
```

```
r2_scorer = make_scorer(r2_score)

kf = KFold(n_splits=10, shuffle=True, random_state=42)

# ------------------------
# METRIC FUNCTION
# ------------------------
def evaluate_model(name, model):
    print(f"\n---------- {name} ----------")

    pipe = Pipeline([
        ("scaler", StandardScaler()),
        ("model", model)
    ])

    r2 = cross_val_score(pipe, X, y, cv=kf, scoring=r2_scorer)
    rmse_val = cross_val_score(pipe, X, y, cv=kf, scoring=rmse_scorer)
    mae = cross_val_score(pipe, X, y, cv=kf, scoring=mae_scorer)

    # DISPLAY ONLY POSITIVE VALUES
    print(f"R² Score: {abs(np.mean(r2)) * 100:.2f} %")
    print(f"RMSE: {abs(np.mean(rmse_val)):.4f}")
    print(f"MAE: {abs(np.mean(mae)):.4f}")


# ------------------------
# ALL MODEL EXECUTION
# ------------------------
evaluate_model("LR = Linear Regression",
               LinearRegression())

evaluate_model("LLR = LASSO Linear Regression",
               Lasso(alpha=0.01))

evaluate_model("RT = Regression Tree",
               DecisionTreeRegressor(random_state=42))

evaluate_model("CT = Conditional Inference Tree",
               DecisionTreeRegressor(criterion="friedman_mse"))

evaluate_model("RF = Random Forest",
               RandomForestRegressor(n_estimators=300, random_state=42))

evaluate_model("XGB = Extreme Gradient Boosting",
               XGBRegressor(n_estimators=500, learning_rate=0.05, max_depth=4,
```

```
                    subsample=0.8, colsample_bytree=0.8, random_state=42,
                    objective='reg:squarederror'))

evaluate_model("KNN = k-Nearest Neighbour",
            KNeighborsRegressor(n_neighbors=5))

evaluate_model("SVM = Support Vector Machine",
            SVR(kernel='rbf', C=10, gamma='scale'))

evaluate_model("NN = Neural Network",
            MLPRegressor(hidden_layer_sizes=(64, 32),
                    activation='relu',
                    max_iter=2000,
                    random_state=42))
```

Python code:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.metrics import (
   cohen_kappa_score, f1_score, roc_auc_score,
   confusion_matrix
)
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline

# MODELS
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier
from xgboost import XGBClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.neural_network import MLPClassifier
from sklearn.naive_bayes import GaussianNB

# -------------------------
# LOAD DATA
# -------------------------
df = pd.read_excel("Nutrition data.xlsx")
```

```
features = ['V106','NV717','V190','NV012','NV212','NV511','NV201',
            'V404','NV312','V406','V405','NV418','NV419','NV467B',
            'NV467D','NV730','V701','NV705','NV467C','Media_access',
            'NV113','NV130','NV136','V025','Family_planning']

X = df[features]
y = df["BMI_Deci_class"]

# -------------------------
# MODEL LIST (ALL VALID)
# -------------------------
models = {
    "LTR": LogisticRegression(max_iter=2000),
    "LLTR": LogisticRegression(penalty='l1', solver='liblinear', max_iter=2000),  # FIXED
    "DT": DecisionTreeClassifier(),
    "RF": RandomForestClassifier(n_estimators=300),
    "CT": DecisionTreeClassifier(criterion='entropy'),
    "ADB": AdaBoostClassifier(),
    "XGB": XGBClassifier(eval_metric='logloss'),
    "KNN": KNeighborsClassifier(n_neighbors=5),
    "SVM": SVC(probability=True),
    "NN": MLPClassifier(hidden_layer_sizes=(64,32), max_iter=2000),
    "NB": GaussianNB()
}

# -------------------------
# EVALUATION FUNCTION
# -------------------------
def evaluate(target_class, label):

    y_binary = (y == target_class).astype(int)  # 1 vs all

    X_train, X_test, y_train, y_test = train_test_split(
        X, y_binary, test_size=0.25, random_state=42
    )

    results = []

    for name, model in models.items():

        pipe = Pipeline([
            ("scaler", StandardScaler()),
            ("model", model)
        ])
```

```
        pipe.fit(X_train, y_train)
        preds = pipe.predict(X_test)
        probs = pipe.predict_proba(X_test)[:,1]

        kappa = cohen_kappa_score(y_test, preds)
        cm = confusion_matrix(y_test, preds)
        specificity = cm[0,0] / cm[0].sum()
        auc = roc_auc_score(y_test, probs)
        f1 = f1_score(y_test, preds)

        results.append([name, kappa, specificity, auc, f1])

    df_results = pd.DataFrame(results, columns=["Model","Kappa","Specificity","AUC","F1"])
    df_results = df_results.set_index("Model")

    # HEATMAP
    plt.figure(figsize=(6,6))
    sns.heatmap(df_results, annot=True, cmap="Greens", fmt=".3f")
    plt.title(f"Prediction Performance — {label}")
    plt.tight_layout()
    plt.show()

# ------------------------
# GENERATE 3 HEATMAPS
# ------------------------
evaluate(1, "Underweight")
evaluate(3, "Overweight")
evaluate(4, "Obese")
```

R code:

```
# ----------------------------
# LOAD LIBRARIES
# ----------------------------
library(ggplot2)
library(reshape2)
library(RColorBrewer)

# ----------------------------
# MODELS
# ----------------------------
models <- c("LTR","LLTR","DT","RF","CT","ADB","XGB","KNN","SVM","NN","NB")
```

```
# ----------------------------
# Example Data (Replace with your real results)
# ----------------------------
set.seed(42)

df_overweight <- data.frame(
  Model = models,
  Specificity = runif(length(models), 0.4, 0.9),
  Cohen_Kappa = runif(length(models), 0.1, 0.7),
  AUC = runif(length(models), 0.5, 0.9),
  F1_score = runif(length(models), 0.5, 0.8)
)

df_obesity <- data.frame(
  Model = models,
  Specificity = runif(length(models), 0.5, 0.95),
  Cohen_Kappa = runif(length(models), 0.3, 0.8),
  AUC = runif(length(models), 0.6, 0.9),
  F1_score = runif(length(models), 0.55, 0.85)
)

# ----------------------------
# FUNCTION TO PLOT HEATMAP
# ----------------------------
plot_heatmap <- function(df, title) {
  df_melt <- melt(df, id.vars = "Model")

  ggplot(df_melt, aes(x = variable, y = Model, fill = value)) +
    geom_tile(color = "white") +
    geom_text(aes(label = sprintf("%.2f", value)), color = "black") +
    scale_fill_gradientn(colors = brewer.pal(9, "Greens"), limits = c(0,1)) +
    labs(title = title, x = "", y = "") +
    theme_minimal(base_size = 12) +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))
}

# ----------------------------
# PLOT HEATMAPS
# ----------------------------
plot_heatmap(df_overweight, "Prediction of the Overweight Class")
```

```
plot_heatmap(df_obesity, "Prediction of the Obesity Class")
```